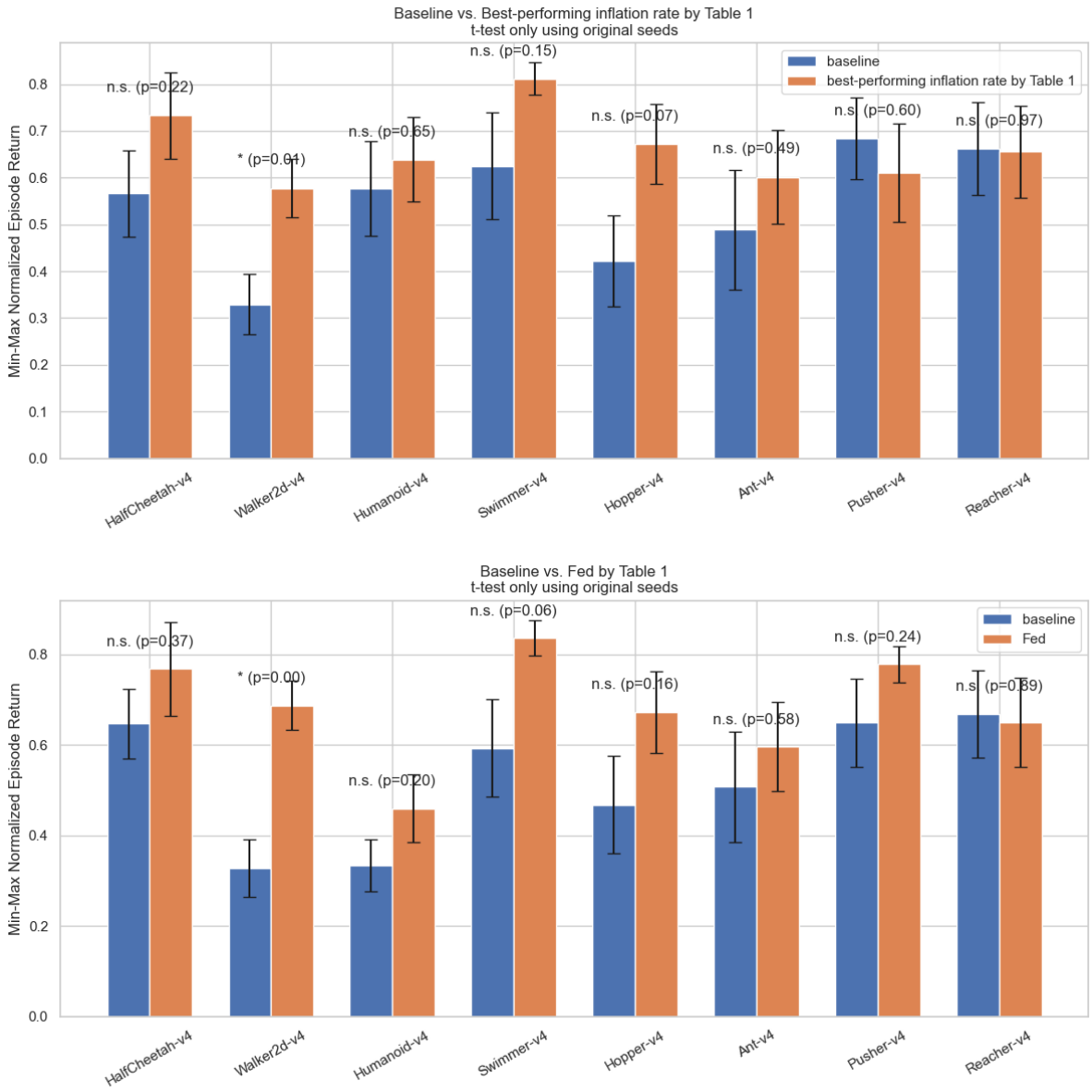

Figure 1: Per-task t-tests across MuJoCo tasks using the original 10 seeds. (Top) Baseline vs. the best-performing inflation rate identified in Table 1 of the main paper. (Bottom) Baseline vs. Fed.

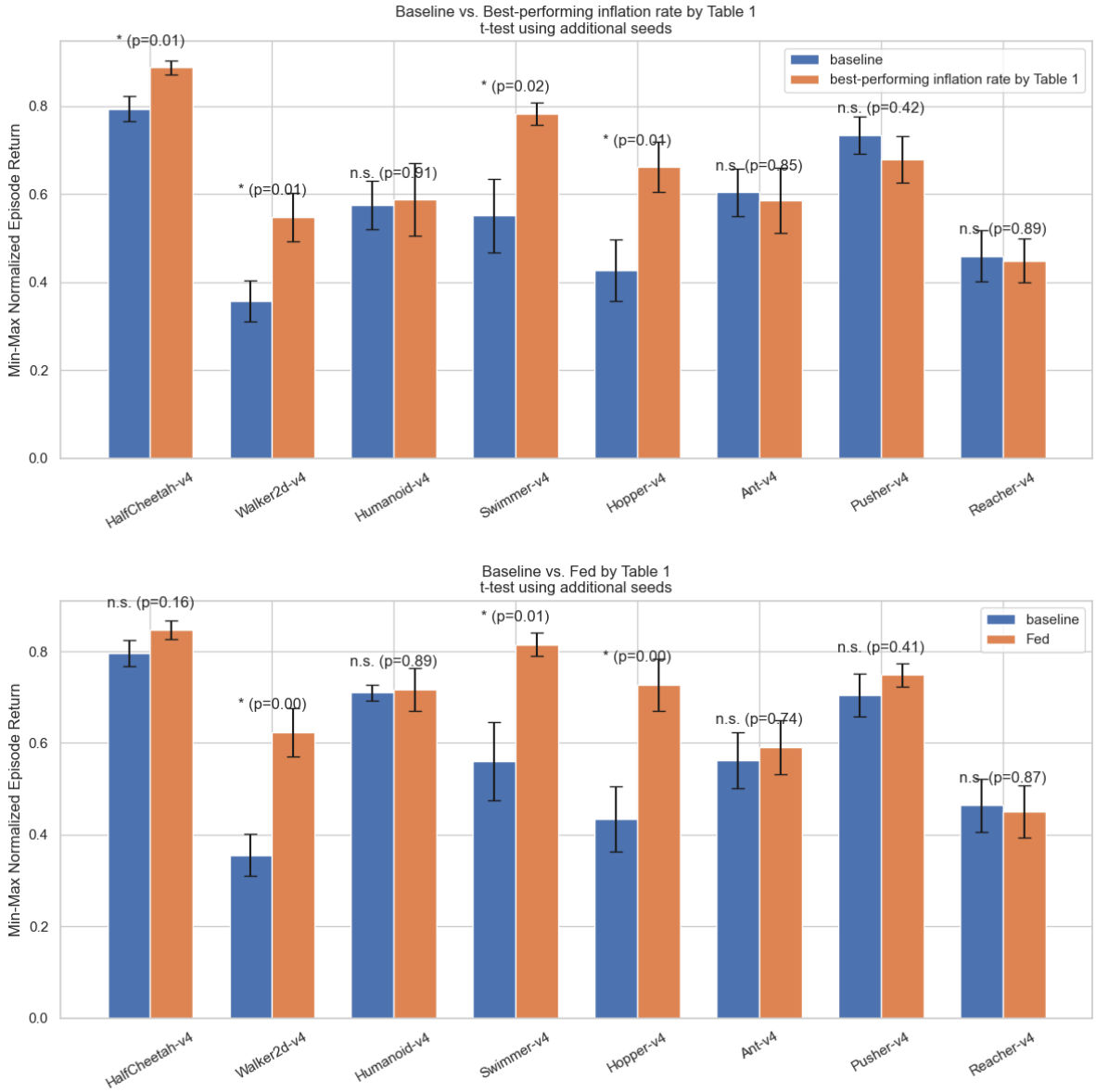

Figure 2: Per-task t-tests across MuJoCo tasks using additional seeds to gain more statistical power. (Top) Baseline vs. the best-performing inflation rate identified in Table 1 of the main paper. (Bottom) Baseline vs. Fed. The total number of seeds, including additional runs, is 42, 18, 18, 18, 18, 26, 26, and 18 for each task, respectively.

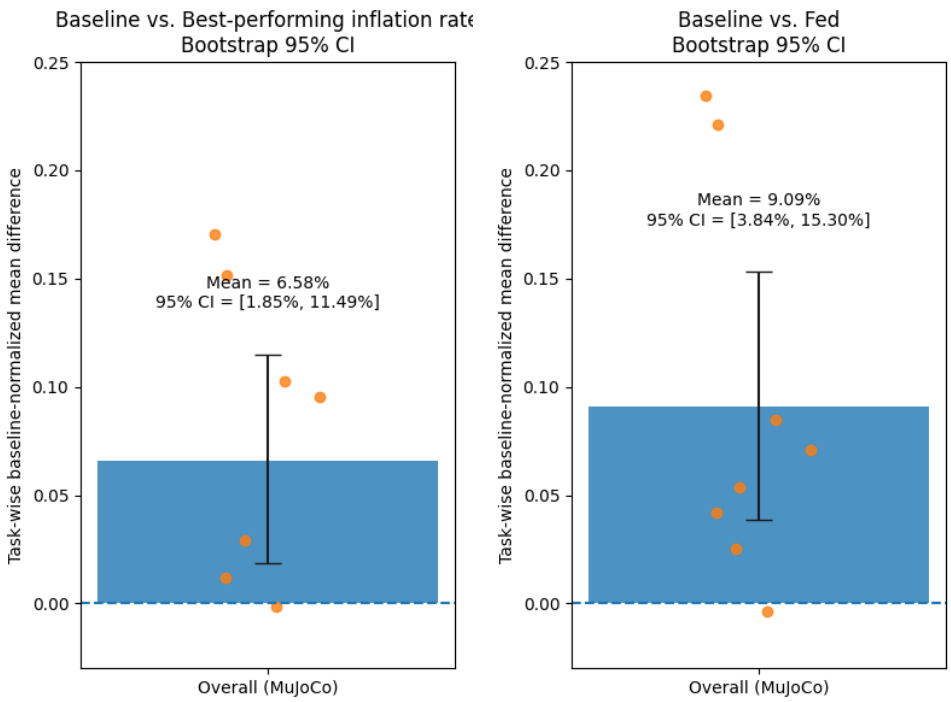

Figure 3: Task-wise bootstrap 95% CI over MuJoCo tasks. (Left) Baseline vs. the best-performing inflation rate identified in Table 1 of the main paper. (Right) Baseline vs. Fed.

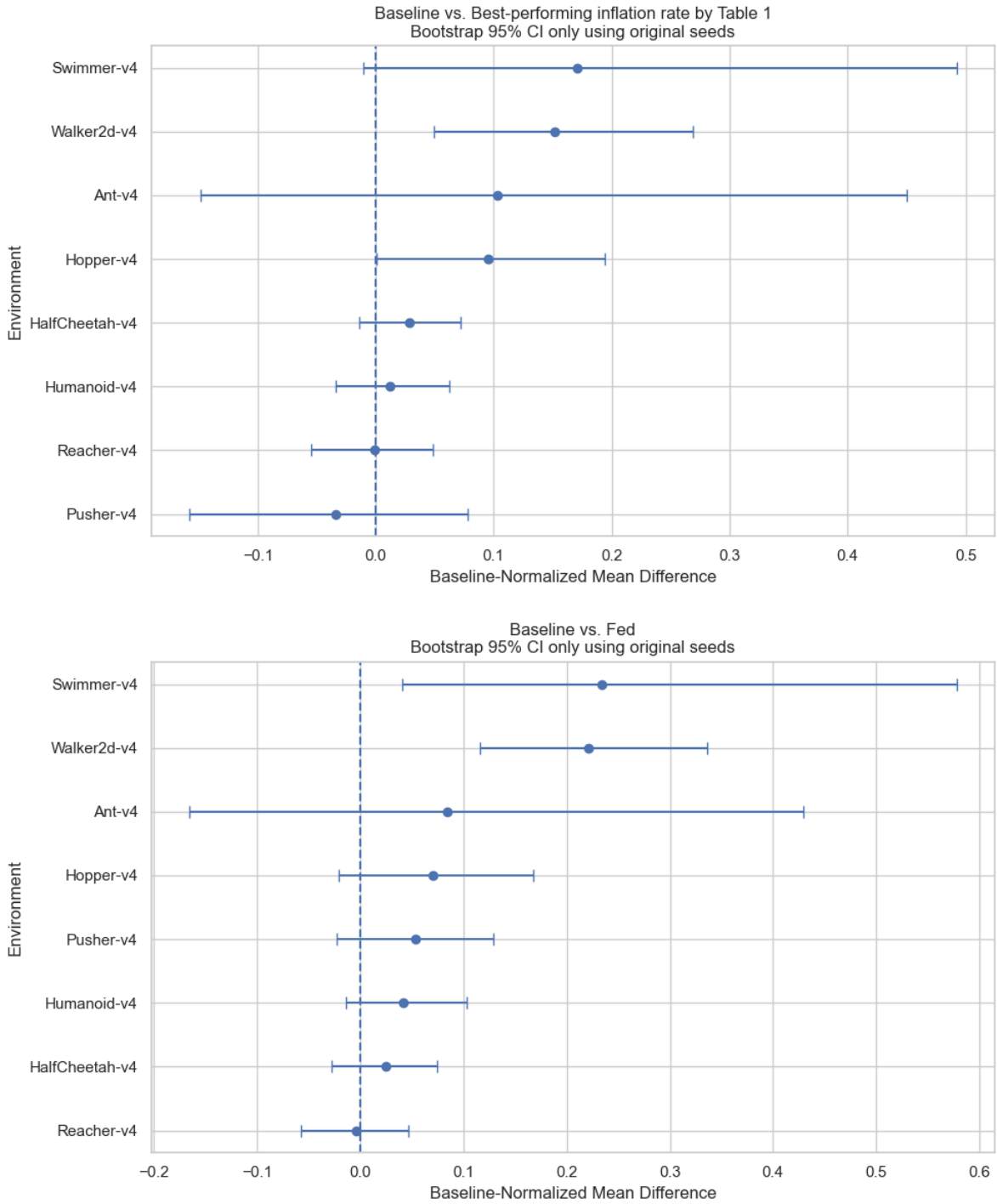

Figure 4: Per-task bootstrap 95% CI across MuJoCo tasks using the original 10 seeds. (Top) Baseline vs. the best-performing inflation rate identified in Table 1 of the main paper. (Bottom) Baseline vs. Fed.

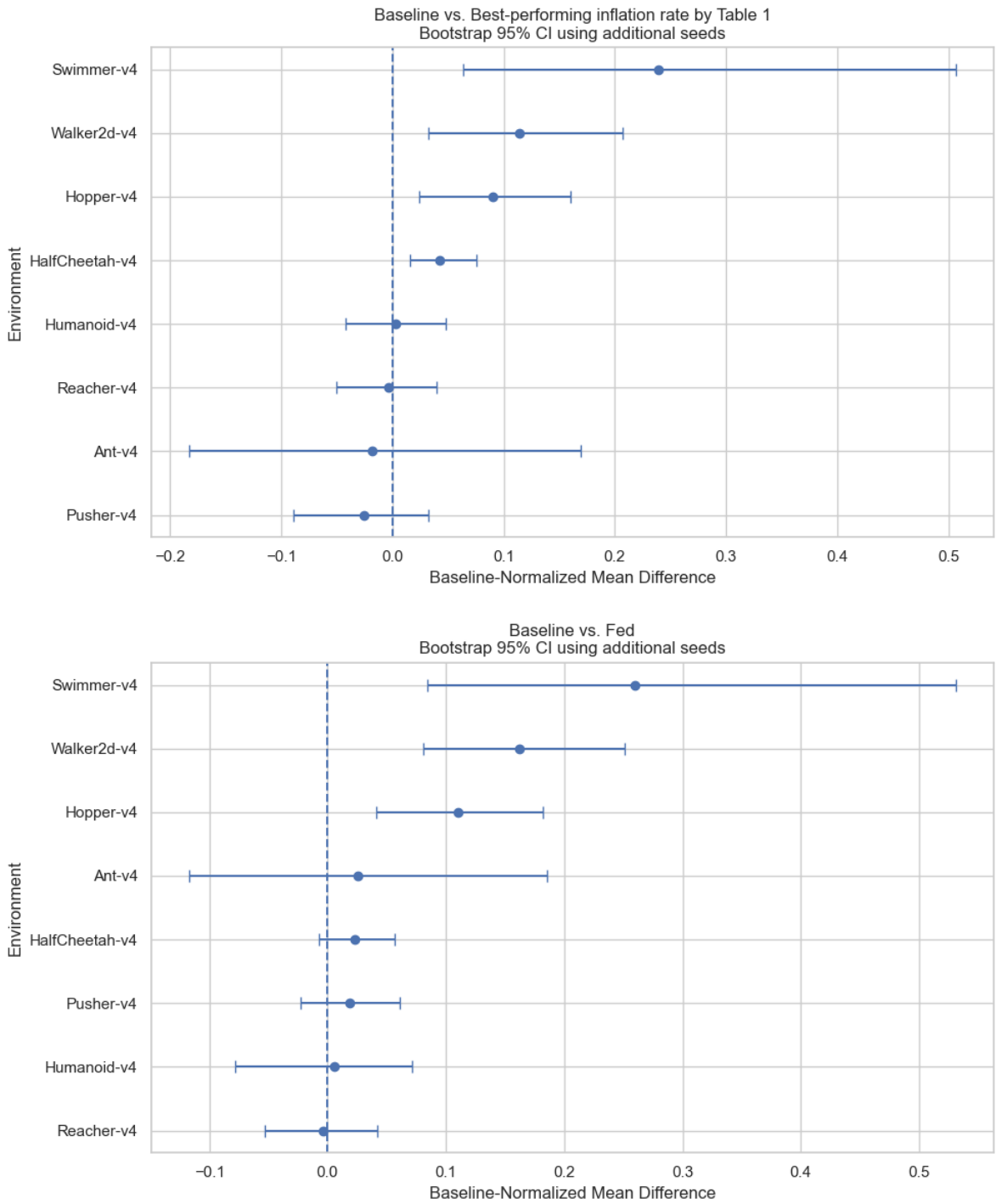

Figure 5: Per-task bootstrap 95% CI across MuJoCo tasks using additional seeds to gain more statistical power. (Top) Baseline vs. the best-performing inflation rate identified in Table 1 of the main paper. (Bottom) Baseline vs. Fed. The total number of seeds, including additional runs, is 42, 18, 18, 18, 18, 26, 26, and 18 for each task, respectively.

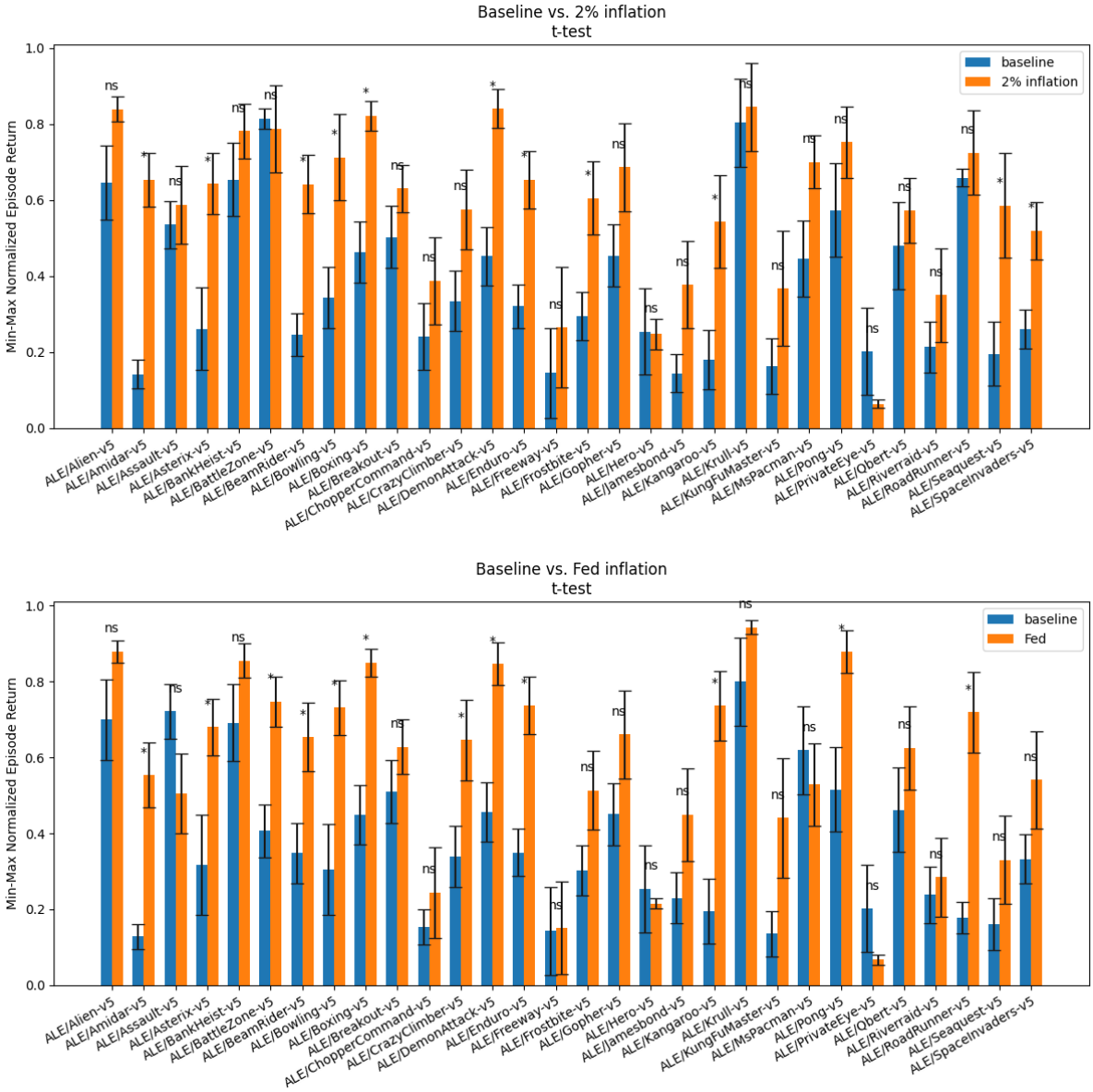

Figure 6: Per-task t-tests across Atari tasks. (Top) Baseline vs. 2% inflation. (Bottom) Baseline vs. Fed.

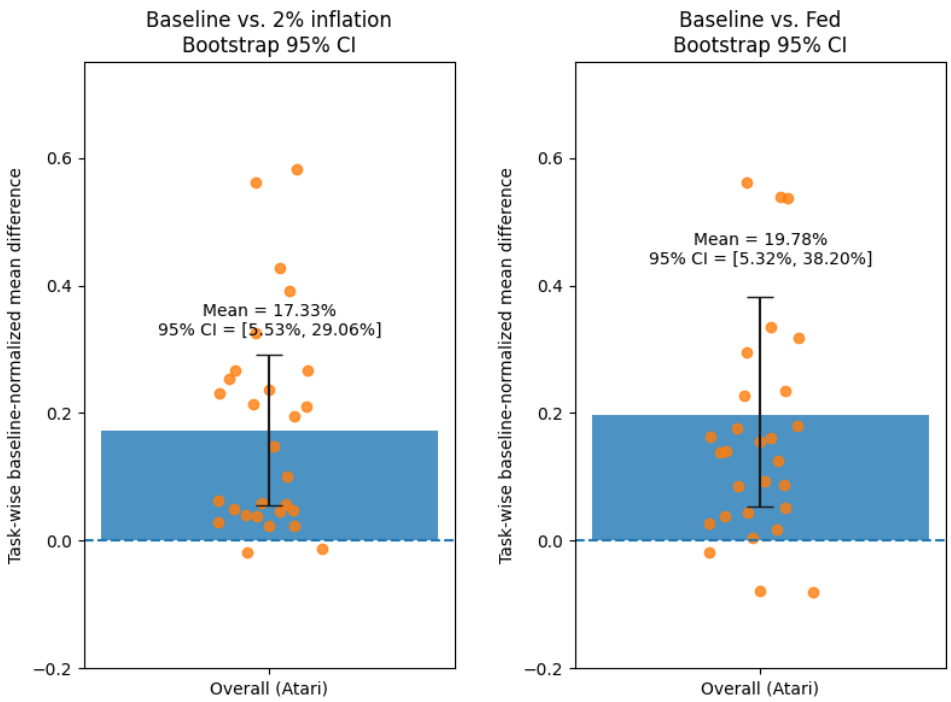

Figure 7: Task-wise bootstrap 95% CI over Atari tasks. (Left) Baseline vs. 2% inflation rate. (Right) Baseline vs. Fed.

Table 1: Performance comparison under different learning rate inflation settings. We added fixed inflation rates (5%, 10%) and the Fed-based adaptive learning rate inflation (lr-Fed) with a 2% target rate.

LR Inflation				
Rate	**HalfCheetah**	**Walker2d**	**Humanoid**	**Swimmer**
0%	**15241±734**	4399±527	**5826±313**	**116±36**
1%	14860±1560	4729±422	5634±151	107±35
2%	14566±1006	**4922±515**	5417±298	77±41
3%	14643±895	4646±660	5282±226	90±41
4%	13854±905	4502±507	5302±280	113±30
5%	12902±1793	4508±472	4952±286	88±36
10%	9932±633	3548±597	2926±476	74±25
lr-Fed (2%)	14783±869	4875±682	5473±408	102±40

Table 2: Performance comparison of baseline (0% reward inflation), 2% reward inflation, and prioritizing recent transition (PRT) in MuJoCo domains.

Reward Inflation vs. Prioritizing Recent Transition (PRT)								
#	**HalfCheetah**	**Walker2d**	**Humanoid**	**Swimmer**	**Hopper**	**Ant**	**Pusher**	**Reacher**
0%	15241±734	4399±527	5826±313	116±36	2869±299	4763±1642	-23.28±2.89	**-3.41±0.23**
2%	**15607±372**	**5102±520**	**5861±199**	**136±11**	**3040±499**	4346±1267	**-22.07±1.58**	-3.42±0.24
PRT	15482±687	5000±546	5730±362	104±40	2923±469	**5466±1318**	-23.32±3.55	-3.43±0.20

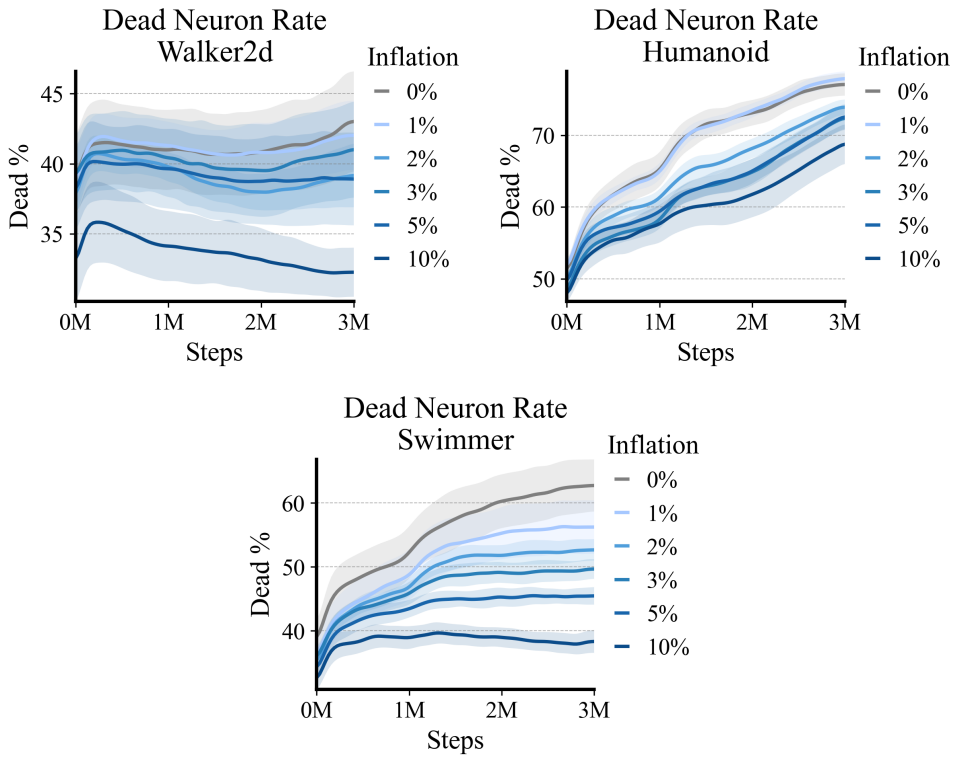

Figure 8: Additional results of dead neuron ratios across different tasks under varying reward inflation rates (Fig. 2(a) in the main paper corresponds to HalfCheetah-v4).

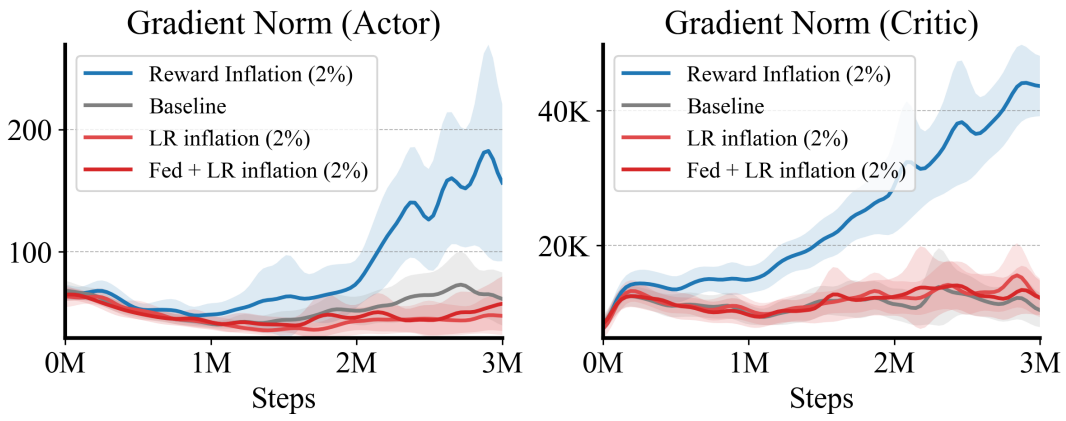

Figure 9: Comparison of gradient norm shift in 2% reward inflation, baseline (no inflation), and 2% learning rate (LR) inflation.

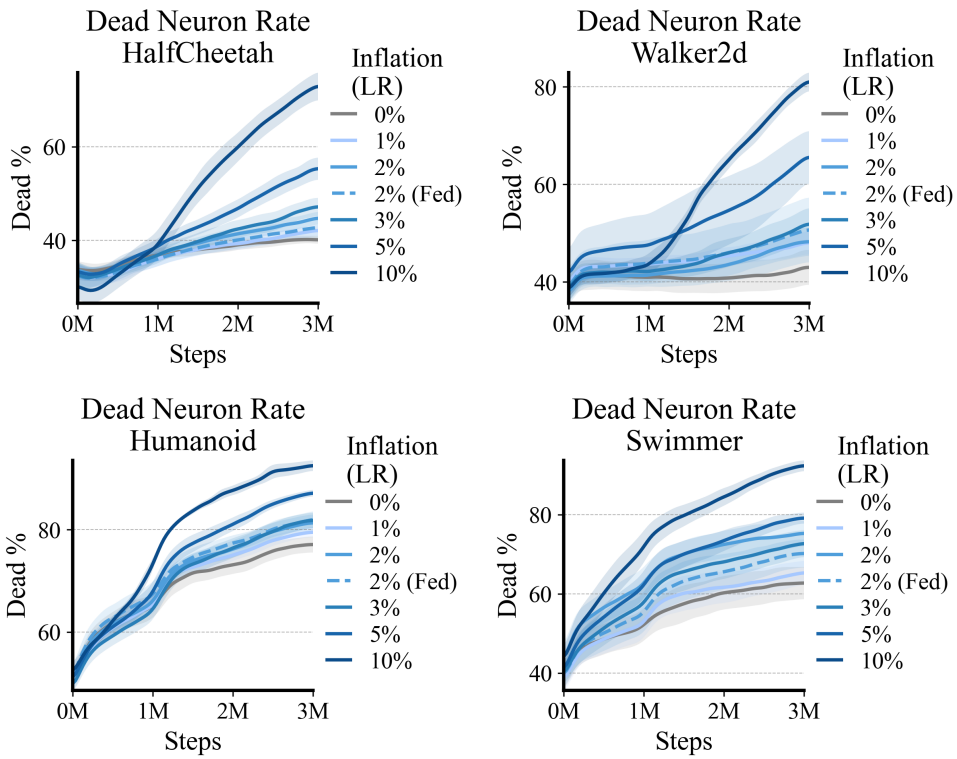

Figure 10: Dead neuron ratios across different task under varying learning rate (LR) inflation rates.

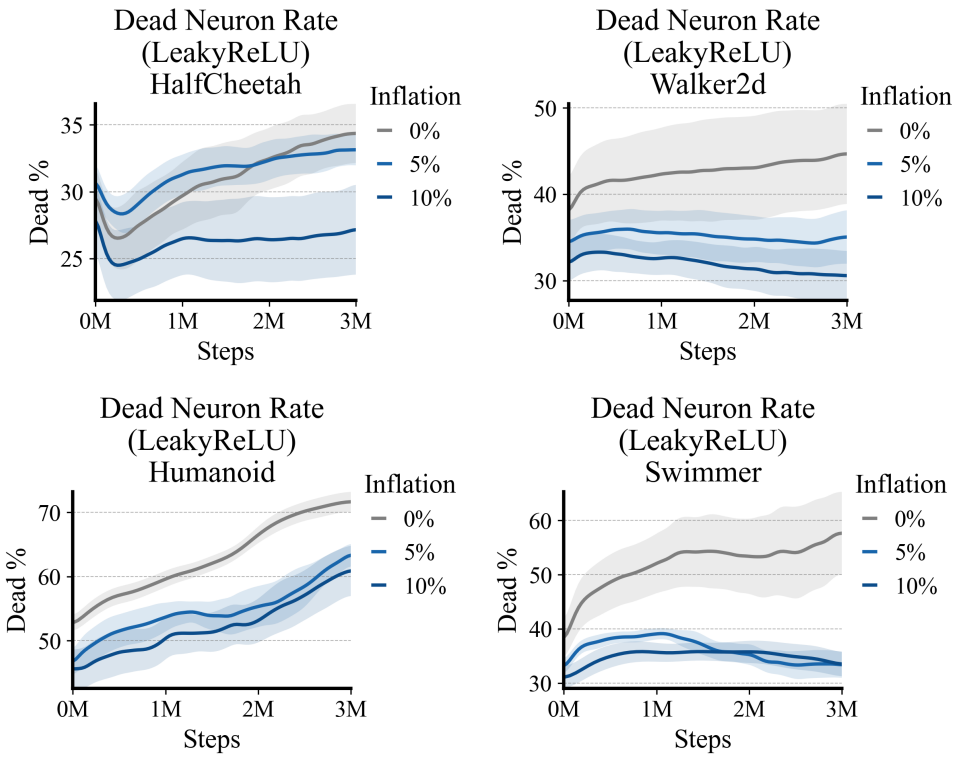

Figure 11: Dead neuron ratios across different tasks under varying reward inflation rates, using `LeakyReLU` instead of `ReLU`.

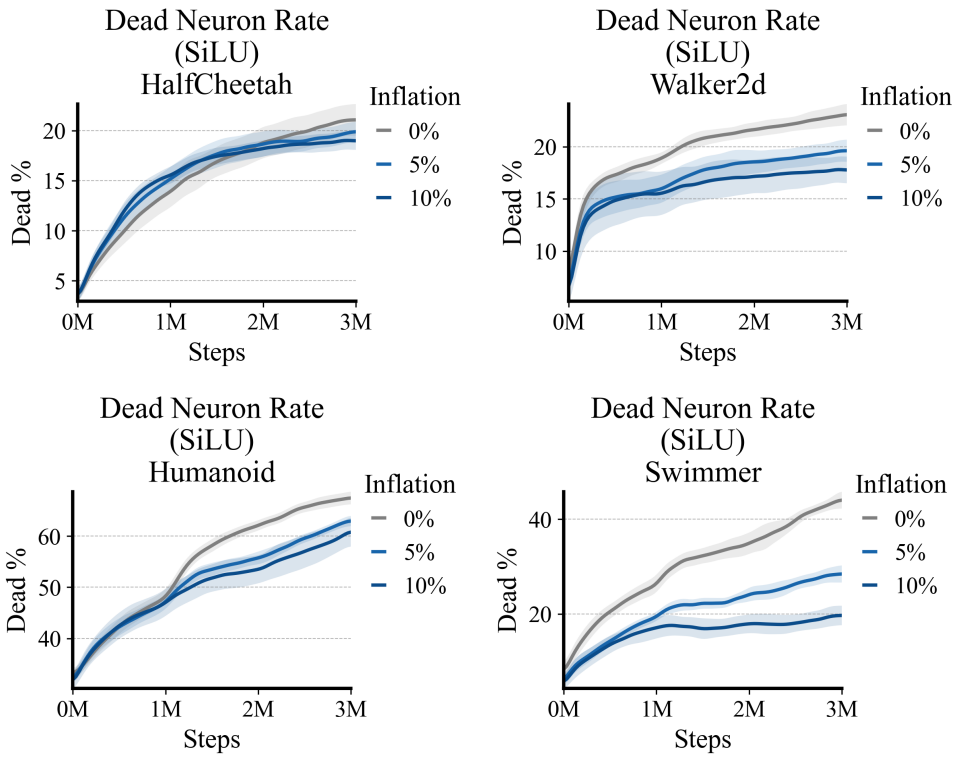

Figure 12: Dead neuron ratios across different tasks under varying reward inflation rates, using `SiLU` instead of `ReLU`.